

SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATIONS

Graduate School of Data Science, Seoul National University

Seoul, Republic of Korea

Integrated M.S. and Ph.D. in Data Science

Sep 2022 – Feb 2027

Advisor: Prof. Sanghack Lee.

Overall GPA: 4.02 /4.3; Ph.D. Qualification Exam partially waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University

Seoul, Republic of Korea

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude*

Mar 2016 – Aug 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.

Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

- ◇ Min Woo Park and Sanghack Lee†. **Counterfactual Structural Causal Bandits**. *International Conference on Learning Representations* (ICLR), 2026.
- ◇ Min Woo Park, Andy Ardit, Elias Bareinboim†, Sanghack Lee†. **Structural Causal Bandits under Markov Equivalence**. *Neural Information Processing Systems* (NeurIPS), 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- ◇ Min Woo Park and Sanghack Lee†. **On Transportability for Structural Causal Bandits**. *arXiv preprint*, 2025. under review.
- ◇ Yonghan Jung†, Min Woo Park, and Sanghack Lee†. **Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference**. *Neural Information Processing Systems* (NeurIPS), 2024.
- ◇ Min Woo Park*, and Ji Eun Lee*†. **Computation of the iterated Aluthge, Duggal, and Mean transforms**. *Filomat*, 2023.
- ◇ Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. **Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery**. under review.
- ◇ Yesong Choe, Yeahoon Kwon, Min Woo Park, and Sanghack Lee†. **Canonical Domain Reduction for Partial Counterfactual Identification**. under review.
- ◇ Ji Eun Lee*† and Min Woo Park*. **Convergence of the iterated mean transforms of a 2×2 matrix**. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship | Seoul National University Spring 2024, Spring 2025, Spring 2026

BK21 scholarship

Fall 2024, Fall 2025

First place merit scholarship | Sejong University

Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University

Nov 2021

▷ Mitigating spurious regression in time-series data via Granger causality.

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University

Nov 2020

▷ The significance of the Euler's number and its applications.

ACADEMIC SERVICES

Conference Reviewer: 2026 UAI, ICLR, ICML, 2025 ICML

PROJECTS

- Towards More Scalable Score-based Causal Discovery** | LG AI Research Sep 2025 – Aug 2026
- ▷ Leading a project on scalable score-based causal discovery algorithms.
(First-author submission under review.)
 - Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
 - Investigating parallelization strategies for score-based causal discovery algorithms.
- Hyundai NGV Data Boot-Campus 2025** | Hyundai Motor July 2025 – Sep 2025
- ▷ Developed high-voltage battery performance prediction technology.
- Hyundai NGV Data Boot-Campus 2024** | Hyundai Motor July 2024 – Sep 2024
- ▷ Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometry-based automatic curvature analysis.
- Deep Reinforcement Learning for Recommendation Systems** | Coursework Project Mar 2024 – Jun 2024
- ▷ Developed a recommendation system based on DDPG, leveraging multi-modal item features (video, audio, and text) to improve recommendation performance.
 - Source code available at github.com/MinwooPark96/ddpgRLRS.
- Optimal Mixed Policies Completeness** | Oxford University Aug 2023 – Feb 2024
- ▷ Studied complete characterization of conditions under which the intervention effect of a variable is positive.
Acknowledged for helpful comments in [Toward a Complete Criterion for Value of Information in Insoluble Decision Problems](#). *Transactions on Machine Learning Research* (TMLR), 2024.

TEACHING ASSISTENTS

- Causal Inference for Data Science Spring 2024, Fall 2024, Spring 2025, Spring 2026
- ▷ Topics included Bayesian networks and graphical causal inference.
- Machine Learning and Deep Learning for Data Science II Fall 2023
- ▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.
- Basic Mathematics for Artificial Intelligence (Undergraduate) Spring 2022
- ▷ Covered linear algebra, basic statistics, and probability theory.

PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

- ICLR 2026 poster session | Rio de Janeiro, Brazil Apr 2026 (scheduled)

Structural Causal Bandits under Markov Equivalence

- ExploreCSR poster session | Seoul, Republic of Korea Dec 2025
- NeurIPS 2025 poster session | San Diego, USA Dec 2025
- JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea Nov 2025
- SNU GSDS student research seminar | Seoul, Republic of Korea Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

- NeurIPS 2024 poster session | Vancouver, Canada Dec 2024
- SNU GSDS student research seminar | Seoul, Republic of Korea Dec 2024

REFERENCE

- Sanghack Lee** | Graduate School of Data Science, Seoul National University sanghack@snu.ac.kr
Associate professor

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University
Associate Professor

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Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University
Associate Professor

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